

CHAPTER 13 VECTOR FUNCTIONS

13.1 Vector Functions and Space Curves

- A **vector-valued function** , or **vector function**, is a function whose domain is a set of real numbers and whose range is a set of vectors.
- For every number t in the domain of $\mathbf{r}$ there is a unique vector denoted by $\mathbf{r}(t)$.
- If $f(t)$, $g(t)$, and $h(t)$ are the components of the vector $\mathbf{r}(t)$, then f , g , and h are real-valued functions called the **component functions** of $\mathbf{r}$ and we can write

$$\mathbf{r}(t) = \langle f(t), g(t), h(t) \rangle = f(t)\mathbf{i} + g(t)\mathbf{j} + h(t)\mathbf{k}$$

- **Example** Given

$$\mathbf{r}(t) = t^3\mathbf{i} + \ln(3-t)\mathbf{j} + \sqrt{t}\mathbf{k}.$$

Find the domain of $\mathbf{r}$.

$$f(t) = t^3, \quad t \in \mathbb{R}$$

$$g(t) = \ln(3-t), \quad 3-t > 0 \Rightarrow t < 3$$

$$h(t) = \sqrt{t}, \quad t \geq 0$$

$$\therefore \text{Domain of } \vec{r} \text{ is } [0, 3).$$

- **Theorem**

Let $\mathbf{r}(t) = f(t)\mathbf{i} + g(t)\mathbf{j} + h(t)\mathbf{k}$. Then $\mathbf{r}$ has limit at $t = a$ if and only if f , g and h have limits at a . In this case,

$$\lim_{t \rightarrow a} \mathbf{r}(t) = \left[\lim_{t \rightarrow a} f(t) \right] \mathbf{i} + \left[\lim_{t \rightarrow a} g(t) \right] \mathbf{j} + \left[\lim_{t \rightarrow a} h(t) \right] \mathbf{k}$$

- **Example** Find $\lim_{t \rightarrow 0} (1 + t^3)\mathbf{i} + te^{-t}\mathbf{j} + \frac{\sin t}{t}\mathbf{k}$.

$$\lim_{t \rightarrow 0} \left\langle 1 + t^3, te^{-t}, \frac{\sin t}{t} \right\rangle$$

$$= \left\langle \lim_{t \rightarrow 0} (1 + t^3), \lim_{t \rightarrow 0} te^{-t}, \lim_{t \rightarrow 0} \frac{\sin t}{t} \right\rangle$$

$$= \langle 1, 0, 1 \rangle$$

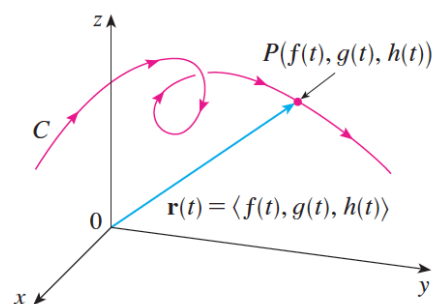
- **Space Curves**

A space curve can be defined parametrically by

$$x = f(t), \quad y = g(t), \quad z = h(t)$$

or written as

$$\mathbf{r}(t) = f(t)\mathbf{i} + g(t)\mathbf{j} + h(t)\mathbf{k}$$



- **Example** Describe the curve defined by the vector function

$$\mathbf{r}(t) = \langle 1 + t, 2 + 5t, -1 + 6t \rangle.$$

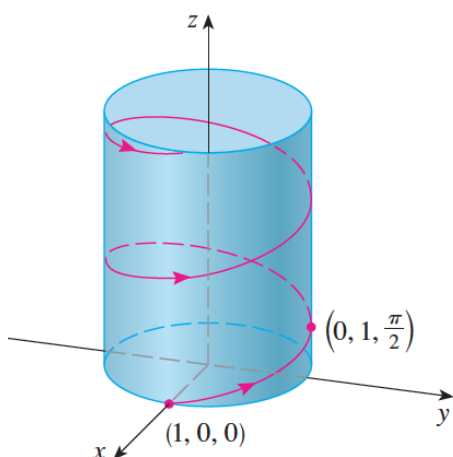
Parametric equations:

$$\begin{aligned} x &= 1 + t \\ y &= 2 + 5t \\ z &= -1 + 6t \end{aligned}$$

A line that passes through point $(1, 2, -1)$ and parallel to the vector $\langle 1, 5, 6 \rangle$.

- **Example** Describe the curve defined by the vector function

$$\mathbf{r}(t) = \cos t \mathbf{i} + \sin t \mathbf{j} + t \mathbf{k}.$$



Parametric equations:

$$\begin{aligned} x &= \cos t \\ y &= \sin t \\ z &= t \end{aligned}$$

$$x^2 + y^2 = \cos^2 t + \sin^2 t = 1$$

$\Rightarrow$ curve lies on the circular cylinder $x^2 + y^2 = 1$.

Circular Helix

$$z = t$$

$\Rightarrow$ curve spirals upward as t increases.

- A line is determined by

- a **fixed point** $P_0(x_0, y_0, z_0)$

- a **fixed direction vector** $\mathbf{v} = a\mathbf{i} + b\mathbf{j} + c\mathbf{k} \approx \langle a, b, c \rangle$

- It is the set of all points P such that

$$\overrightarrow{P_0P} = t\mathbf{v}$$

for some real number t .

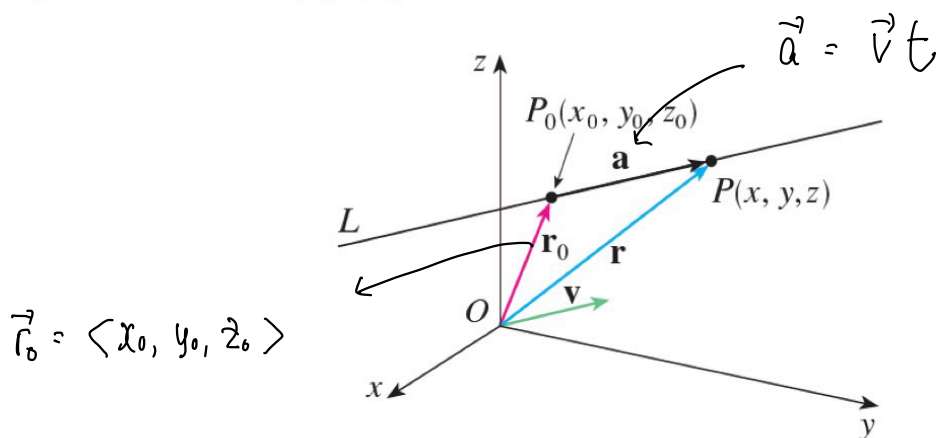
- If $\mathbf{r} = \overrightarrow{OP}$ and $\mathbf{r}_0 = \overrightarrow{OP_0}$, then

$$\mathbf{r} = \mathbf{r}_0 + t\mathbf{v}.$$

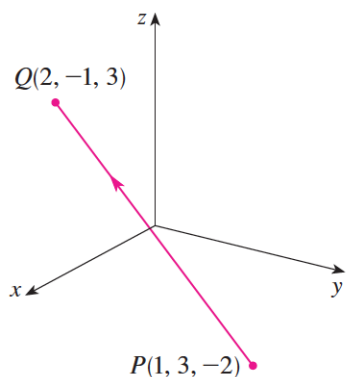
- For $\mathbf{r} = \langle x, y, z \rangle$, $\mathbf{r}_0 = \langle x_0, y_0, z_0 \rangle$ and $\mathbf{v} = \langle a, b, c \rangle$, we have

$$x = x_0 + at, \quad y = y_0 + bt, \quad z = z_0 + ct$$

- These are the **parametric equations** of the line through the point $P_0(x_0, y_0, z_0)$ and parallel to $\mathbf{v} = \langle a, b, c \rangle$.



- **Example** Find a vector equation and parametric equations for the line segment that joins the point $P(1, 3, -2)$ and $Q(2, -1, 3)$.



Point : $(1, 3, -2)$

$\vec{OQ} - \vec{OP}$

Direction : $\vec{PQ} = \langle 2-1, -1-3, 3-(-2) \rangle$
 $= \langle 1, -4, 5 \rangle$

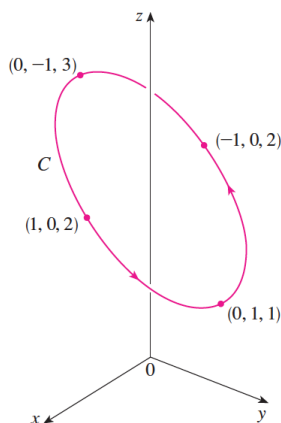
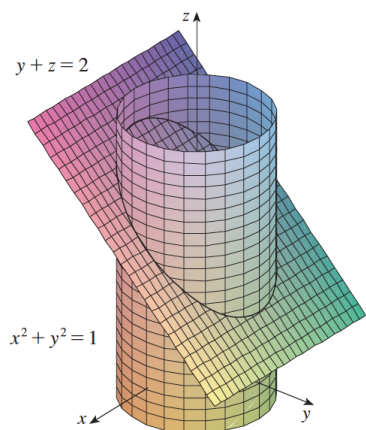
vector
function

$\vec{r}(t) = \langle 1+t, 3-4t, -2+5t \rangle$

Parametric equations : $x = 1+t, y = 3-4t, z = -2+5t$

$0 \leq t \leq 1$

- **Example** Find a vector function that represents the curve of intersection of the cylinder $x^2 + y^2 = 1$ and the plane $y + z = 2$.



$x^2 + y^2 = 1$

$\Rightarrow x = \cos t, y = \sin t$

$0 \leq t \leq 2\pi$

$y + z = 2$

$\Rightarrow z = 2 - y = 2 - \sin t$

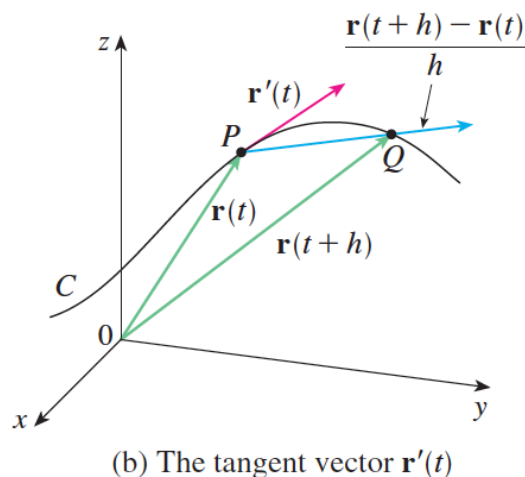
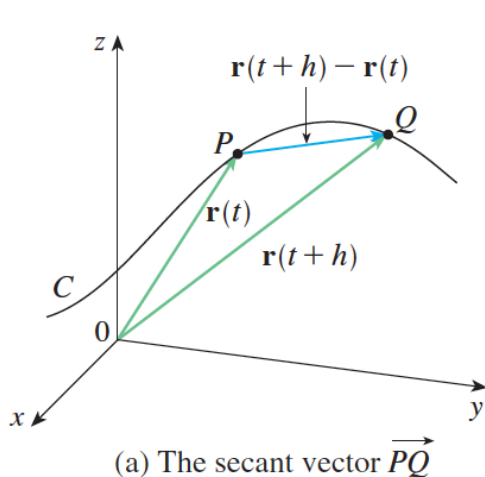
$\therefore \vec{r}(t) = \langle \cos t, \sin t, 2 - \sin t \rangle, 0 \leq t \leq 2\pi.$

Exercises Page 853 1, 3, 5, 17, 21, 23, 25, 27, 43 (notice that $z = \sqrt{x^2 + y^2} = 1 + y$).

13.2 Derivatives and Integrals of Vector Functions

- **Derivatives** The derivative $\mathbf{r}'$ of a vector function $\mathbf{r}$ is defined as:

$$\frac{d\mathbf{r}}{dt} = \mathbf{r}'(t) = \lim_{h \rightarrow 0} \frac{\mathbf{r}(t+h) - \mathbf{r}(t)}{h}$$



- $\mathbf{r}'(t)$ is the **tangent vector** to the curve defined by $\mathbf{r}$ at point P .
- The **unit** tangent vector is

$$\mathbf{T}(t) = \frac{\mathbf{r}'(t)}{\|\mathbf{r}'(t)\|} \quad \text{length}$$

- **Theorem**

If $\mathbf{r}(t) = f(t)\mathbf{i} + g(t)\mathbf{j} + h(t)\mathbf{k}$, where f , g , and h are differentiable functions, then

$$\mathbf{r}'(t) = f'(t)\mathbf{i} + g'(t)\mathbf{j} + h'(t)\mathbf{k}$$

- **Example** If

$$\mathbf{r}(t) = (t^2 + t)\mathbf{i} + e^t\mathbf{j} + 2\mathbf{k},$$

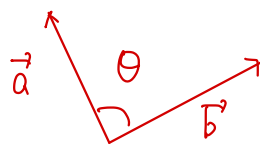
find $\mathbf{r}'(t)$, $\mathbf{r}''(t)$, and the angle between $\mathbf{r}'(0)$ and $\mathbf{r}''(0)$.

$$\vec{r}(t) = \langle t^2 + t, e^t, 2 \rangle$$

$$\vec{r}'(t) = \langle 2t + 1, e^t, 0 \rangle$$

$$\vec{r}''(t) = \langle 2, e^t, 0 \rangle$$

$$\vec{a} \cdot \vec{b} = \|\vec{a}\| \|\vec{b}\| \cos \theta$$



$$\vec{r}'(0) = \langle 1, 1, 0 \rangle$$

$$\vec{r}''(0) = \langle 2, 1, 0 \rangle$$

$$\|\vec{r}'(0)\| = \sqrt{1^2 + 1^2 + 0^2} = \sqrt{2}$$

$$\|\vec{r}''(0)\| = \sqrt{2^2 + 1^2 + 0^2} = \sqrt{5}$$

$$\vec{r}'(0) \cdot \vec{r}''(0) = \|\vec{r}'(0)\| \|\vec{r}''(0)\| \cos \theta$$

$$(1)(2) + (1)(1) + (0)(0) = (\sqrt{2})(\sqrt{5}) \cos \theta$$

$$\cos \theta = \frac{3}{\sqrt{10}}$$

$$\theta = 0.32 \text{ rad.} \quad \text{or} \quad 18.4^\circ$$

- **Example** Find parametric equations for the tangent line to the curve determined by

$$\mathbf{r}(t) = 2 \cos t \mathbf{i} + \sin t \mathbf{j} + t \mathbf{k}$$

at $(0, 1, \pi/2)$.

$$\vec{r}'(t) = \langle -2 \sin t, \cos t, 1 \rangle$$

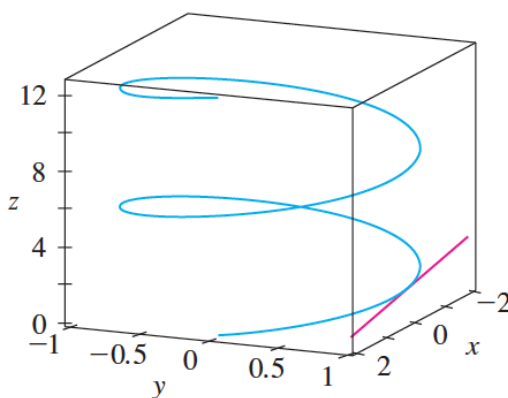
At the point,

$$\begin{cases} 2 \cos t = 0 \\ \sin t = 1 \\ t = \frac{\pi}{2} \end{cases} \Rightarrow t = \frac{\pi}{2}$$

To get the tangent line:

Point : $(0, 1, \frac{\pi}{2})$

Direction : $\vec{r}'(\frac{\pi}{2}) = \langle -2, 0, 1 \rangle$



Parametric equations

$$x = -2t$$

$$y = 1$$

$$z = \frac{\pi}{2} + t$$

$$t \in \mathbb{R}$$

- **Differentiation Rules**

Let $\mathbf{u}$ and $\mathbf{v}$ be differentiable, vector-valued functions, f a differentiable, real-valued function, and c a scalar. Then

1. $\frac{d}{dt} [\mathbf{u}(t) \pm \mathbf{v}(t)] = \mathbf{u}'(t) \pm \mathbf{v}'(t)$

2. $\frac{d}{dt} [c\mathbf{u}(t)] = c\mathbf{u}'(t)$

3. $\frac{d}{dt} [f(t)\mathbf{u}(t)] = f(t)\mathbf{u}'(t) + f'(t)\mathbf{u}(t)$

4. $\frac{d}{dt} [\mathbf{u}(t) \cdot \mathbf{v}(t)] = \mathbf{u}'(t) \cdot \mathbf{v}(t) + \mathbf{u}(t) \cdot \mathbf{v}'(t)$

5. $\frac{d}{dt} [\mathbf{u}(t) \times \mathbf{v}(t)] = \mathbf{u}'(t) \times \mathbf{v}(t) + \mathbf{u}(t) \times \mathbf{v}'(t)$

6. $\frac{d}{dt} [\mathbf{u}(f(t))] = \mathbf{u}'(f(t))f'(t)$

- **Example** Show that if $\|\mathbf{r}(t)\| = C$ (a constant), then $\mathbf{r}'(t)$ is perpendicular to $\mathbf{r}(t)$ for all t .

$$\|\vec{r}(t)\|^2 = C^2$$

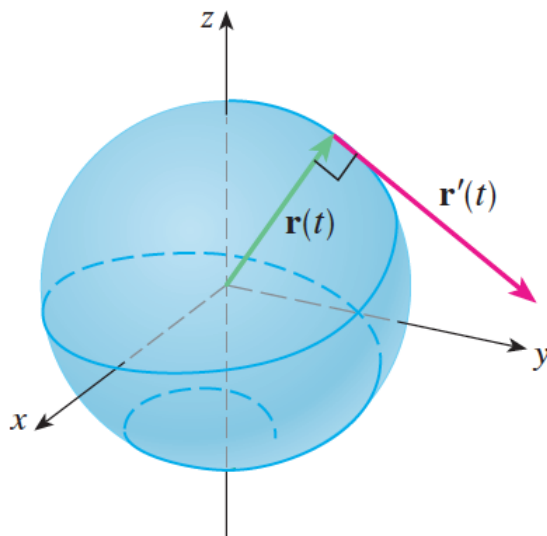
$$\frac{d}{dt} [\vec{r}(t) \cdot \vec{r}(t)] = \frac{d}{dt} C^2$$

$$\vec{r}(t) \cdot \vec{r}'(t) + \vec{r}'(t) \cdot \vec{r}(t) = 0$$

$$2 \vec{r}(t) \cdot \vec{r}'(t) = 0$$

$$\vec{r}(t) \cdot \vec{r}'(t) = 0$$

$\therefore \vec{r}(t)$ and $\vec{r}'(t)$ are perpendicular.



- **Integrals** The definite integral of a continuous vector function $\mathbf{r}(t)$ can be defined as

If $\mathbf{r}(t) = f(t)\mathbf{i} + g(t)\mathbf{j} + h(t)\mathbf{k}$, then

$$\begin{aligned}\int_a^b \mathbf{r}(t) dt &= \lim_{n \rightarrow \infty} \sum_{i=1}^n \mathbf{r}(t_i^*) \Delta t \\ &= \left[\int_a^b f(t) dt \right] \mathbf{i} + \left[\int_a^b g(t) dt \right] \mathbf{j} + \left[\int_a^b h(t) dt \right] \mathbf{k}\end{aligned}$$

- **Example** Given $\mathbf{r}(t) = (t^2 + t)\mathbf{i} + e^{-t}\mathbf{j} - 2\mathbf{k}$, find

$$\begin{aligned}\text{(a)} \quad \frac{d}{dt} [t^3 \mathbf{r}(t)] & \quad t^3 \langle t^2 + t, e^{-t}, -2 \rangle \\ &= \langle \frac{d}{dt}(t^5 + t^4), \frac{d}{dt} t^3 e^{-t}, \frac{d}{dt} (-2t^3) \rangle = \langle t^3(t^2 + t), t^3 e^{-t}, t^3(-2) \rangle \\ &= \langle 5t^4 + 4t^3, 3t^2 e^{-t} - t^3 e^{-t}, -6t^3 \rangle\end{aligned}$$

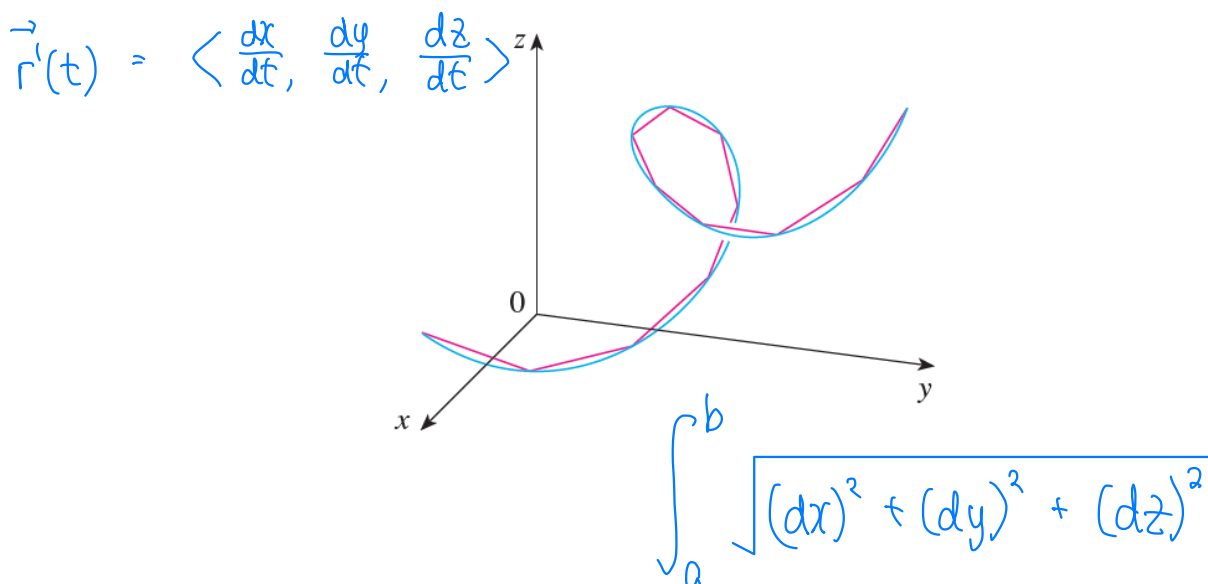
$$\begin{aligned}\text{(b)} \quad \int_0^1 \mathbf{r}(t) dt &= \langle \int_0^1 (t^2 + t) dt, \int_0^1 e^{-t} dt, \int_0^1 (-2) dt \rangle \\ &= \langle \left[\frac{t^3}{3} + \frac{t^2}{2} \right]_0^1, [-e^{-t}]_0^1, [-2t]_0^1 \rangle \\ &= \langle \frac{5}{6}, 1 - e^{-1}, -2 \rangle\end{aligned}$$

13.3 Arc Length

- **Length of a Curve** The length of a space curve is defined as

Suppose the curve has the vector equation $\mathbf{r}(t) = x(t)\mathbf{i} + y(t)\mathbf{j} + z(t)\mathbf{k}$, $a \leq t \leq b$, then

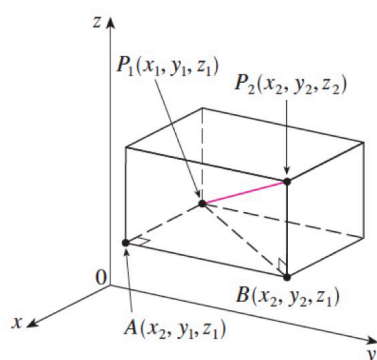
$$\begin{aligned} L &= \int_a^b \sqrt{\left(\frac{dx}{dt}\right)^2 + \left(\frac{dy}{dt}\right)^2 + \left(\frac{dz}{dt}\right)^2} dt \\ &= \int_a^b \|\mathbf{r}'(t)\| dt. \end{aligned}$$



- **Distance Formula in Three Dimensions**

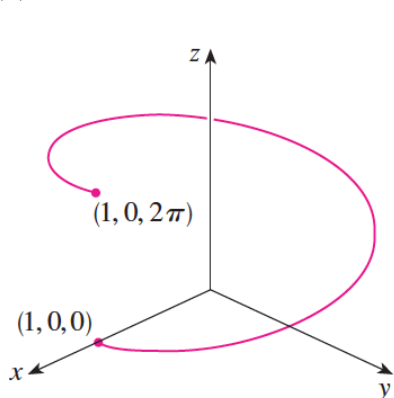
The distance between the points $P_1(x_1, y_1, z_1)$ and $P_2(x_2, y_2, z_2)$ is

$$|P_1P_2| = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2 + (z_2 - z_1)^2}$$



$$\int_a^b \sqrt{\left(\frac{dx}{dt}\right)^2 + \left(\frac{dy}{dt}\right)^2 + \left(\frac{dz}{dt}\right)^2} dt$$

- **Example** Find the length of the arc of the circular helix with vector equation $\mathbf{r}(t) = \cos t \mathbf{i} + \sin t \mathbf{j} + t \mathbf{k}$ from the point $(1, 0, 0)$ to the point $(1, 0, 2\pi)$.



$$0 \leq t \leq 2\pi$$

$$x(t) = \cos t$$

$$\frac{dx}{dt} = -\sin t$$

$$y(t) = \sin t$$

$$\frac{dy}{dt} = \cos t$$

$$z(t) = t$$

$$\frac{dz}{dt} = 1$$

$$L = \int_0^{2\pi} \sqrt{(-\sin t)^2 + (\cos t)^2 + 1^2} dt$$

$$= \int_0^{2\pi} \sqrt{2} dt$$

$$= \sqrt{2} [t]_0^{2\pi}$$

$$= 2\sqrt{2} \pi \quad (\text{units})$$

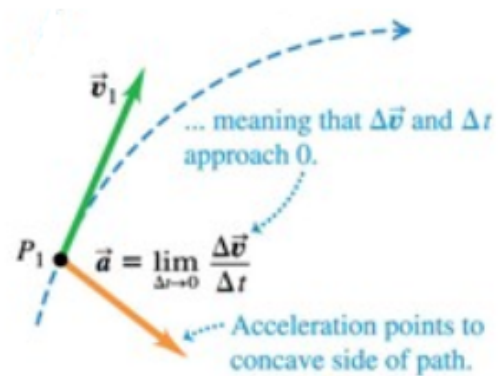
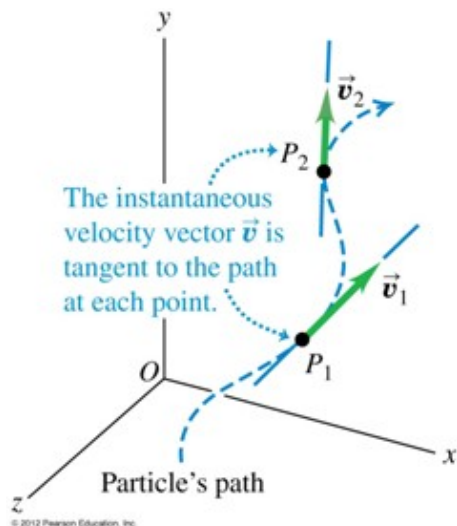
13.4 Motion in Space: Velocity and Acceleration

- Curvilinear Motion** The velocity vector $\mathbf{v}(t)$ and acceleration vector $\mathbf{a}(t)$ of the moving point P with position vector $\mathbf{r}(t) = f(t)\mathbf{i} + g(t)\mathbf{j} + h(t)\mathbf{k}$ are given respectively by

$$\mathbf{v}(t) = \mathbf{r}'(t) = f'(t)\mathbf{i} + g'(t)\mathbf{j} + h'(t)\mathbf{k}$$

$$\mathbf{a}(t) = \mathbf{r}''(t) = f''(t)\mathbf{i} + g''(t)\mathbf{j} + h''(t)\mathbf{k}$$

- The acceleration vector points to the concave side of the curve.



- The speed is

$$v = \|\mathbf{v}(t)\| = \|\mathbf{r}'(t)\|$$

- **Example** The position vector of an object moving in a plane is given by

$$\mathbf{r}(t) = t^3\mathbf{i} + t^2\mathbf{j}.$$

Find its velocity, speed, and acceleration when $t = 1$.

$$\mathbf{r}(t) = \langle t^3, t^2 \rangle$$

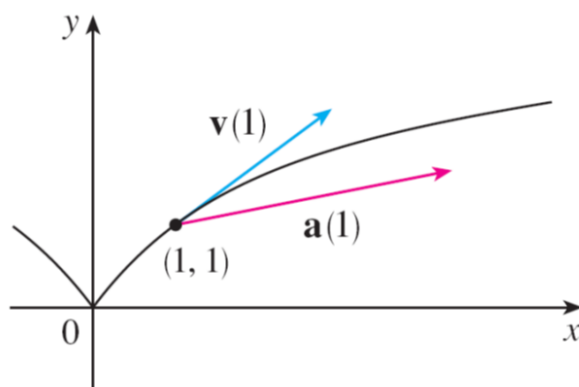
$$\vec{r}'(t) = \langle 3t^2, 2t \rangle$$

$$\vec{r}'(1) = \langle 3, 2 \rangle$$

$$\|\vec{r}'(1)\| = \sqrt{3^2 + 2^2} = \sqrt{13}$$

$$\vec{r}''(t) = \langle 6t, 2 \rangle$$

$$\vec{r}''(1) = \langle 6, 2 \rangle$$



- **Example** A moving particle starts at an initial position $\mathbf{r}(0) = \langle 1, 0, 0 \rangle$ with initial velocity $\mathbf{v}(0) = \mathbf{i} - \mathbf{j} + \mathbf{k}$. Its acceleration is $\mathbf{a}(t) = 4t\mathbf{i} + 6t\mathbf{j} + \mathbf{k}$. Find its velocity and position at time t .

$$\vec{v}(t) = ? \quad \vec{r}(t) = ?$$

$$\vec{a}(t) = \langle 4t, 6t, 1 \rangle$$

$$\vec{v}(t) = \left\langle \frac{4t^2}{2} + C_1, \frac{6t^2}{2} + C_2, t + C_3 \right\rangle$$

$$\vec{v}(0) = \langle C_1, C_2, C_3 \rangle = \langle 1, -1, 1 \rangle$$

$$\Rightarrow \vec{v}(t) = \langle 2t^2 + 1, 3t^2 - 1, t + 1 \rangle$$

$$\vec{r}(t) = \left\langle \frac{2}{3}t^3 + t + C_4, t^3 - t + C_5, \frac{1}{2}t^2 + t + C_6 \right\rangle$$

$$\vec{r}(0) = \langle C_4, C_5, C_6 \rangle = \langle 1, 0, 0 \rangle$$

$$\Rightarrow \vec{r}(t) = \left\langle \frac{2}{3}t^3 + t + 1, t^3 - t, \frac{1}{2}t^2 + t \right\rangle$$